Started on	Saturday, 10 October 2026, 1:44 PM
State	Finished
Completed on	Saturday, 10 October 2026, 2:10 PM
Time taken	26 mins 8 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int main()
{
    int n,a[100],dp[100],max=1;

    scanf("%d",&n);


    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }

    for (int i=0;i<n;i++){
        dp[i]=1;

        for(int j=0;j<i;j++){
            if(a[j] <= a[i] && dp[j]+1 > dp[i])
                dp[i]=dp[j]+1;
        }
        if(dp[i]>max)
           max=dp[i];
    }
    printf("%d",max);
    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.